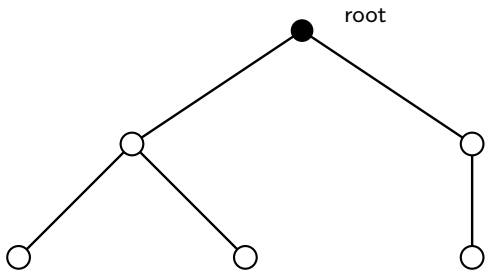
Week 12 / Catalan bijections / Grades 4–5

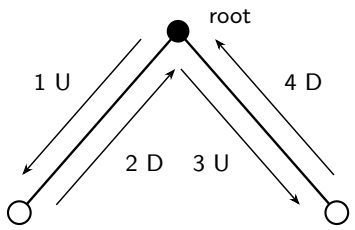
Draw dots joined by lines called edges. The filled dot at the top is the root. Each edge joins a parent dot to a child dot below it. Every dot except the root has exactly one parent. Keep each parent's children in a fixed left-to-right order. All dots connect to the root. Only the branching and its order matter.

Example with five edges:

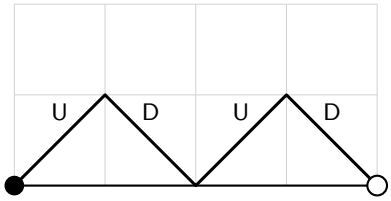


Start and finish at the root. Visit each child's whole branch from left to right before visiting the next child. Write U whenever you go away from the root along an edge and D whenever you return along it.

Example:



On the path, U goes up and D goes down.



Walk code: `UDUD`

Problem 3: Draw a tree for each code. Can a code describe two different trees?

UUDDUD



UDUUDD



UUDUDUDD



UUUDDUDD



Problem 4: Which of these codes could come from a tree? Give rules that decide for any string of U's and D's, and explain why they work. Describe how to rebuild a tree from any code that follows your rules.

UUUUDDDD

UUDDDUUD

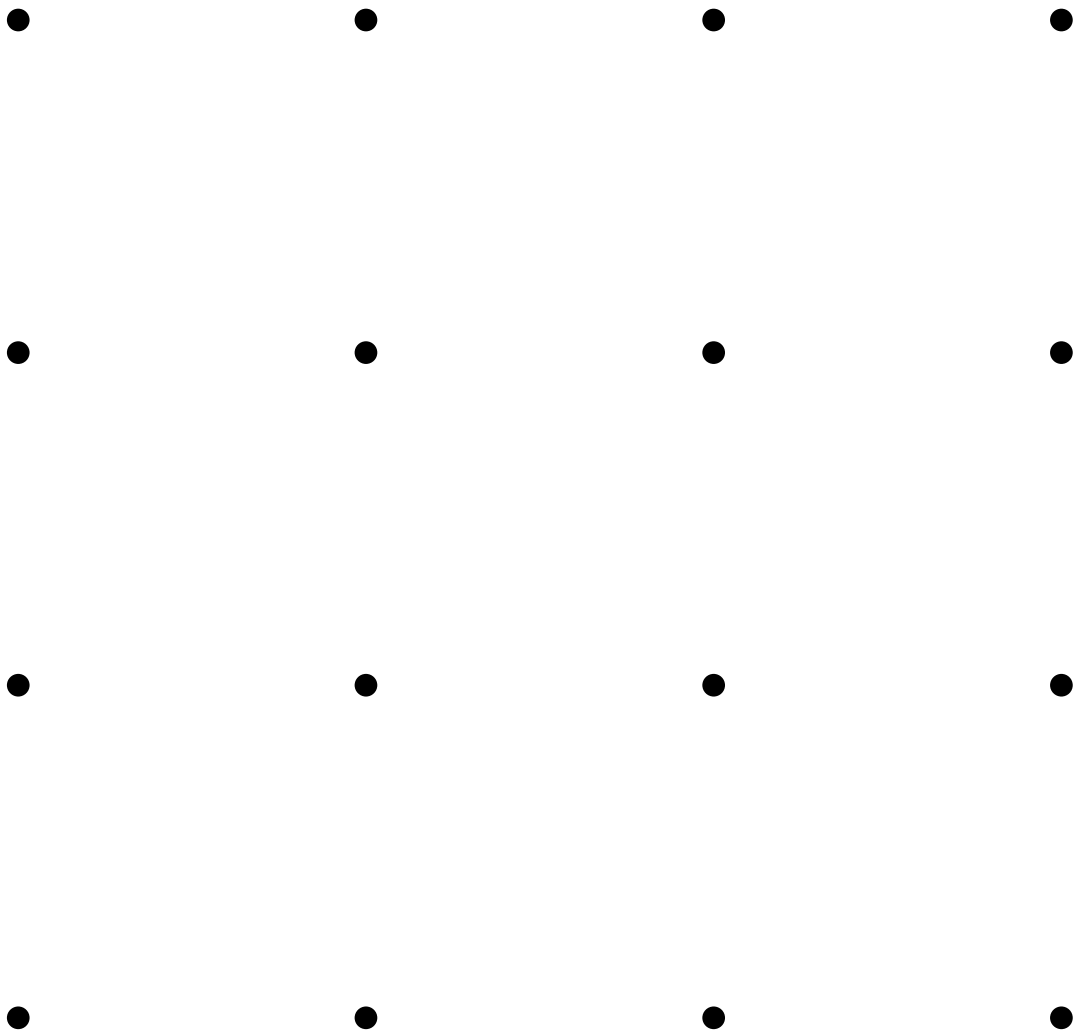
UDUDUUDD

UUDUDUDU

UUUDUDDD

DUUUDDUD

Problem 5: How many different trees have four edges? Organize the count so someone can check that every tree is counted exactly once.



Problem 6: Find the number of trees with five edges, then six edges. Give a counting rule that works for any number of edges, and explain why it counts each tree once.